

Difference-Aware Multiscale Feature Aggregation Network for Building Change Detection

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Abstract—The application of remote sensing (RS) for building change detection (BCD) is indispensable for assessing shifts in land use and surface dynamics. Nevertheless, off-the-shelf deep learning (DL)-based BCD methods often suffer from incomplete change boundaries and pseudo changes, due to the insufficient utilization of difference information from bitemporal RS images. To address these issues, we propose a difference-aware multiscale feature aggregation network (DMFANet) aimed at investigating more representative forms of change representation in BCD tasks. To facilitate comprehensive bitemporal feature alignment and differencing, the feature modulation module (FMM) is designed, which focuses on developing semantically robust and contextually enriched pyramidal feature representations by channel-spatial modulation. Subsequently, the cross-domain difference enhancement module (CDEM) is introduced to accurately locate changed building areas with intricate details by capturing multiperspective difference information through the construction of different domain dependencies. Moreover, we propose a multiscale context aggregation module (MCAM) to effectively adapt to building scale variations by aggregating multiscale difference features under the guidance of contextual information while mitigating the interference of redundant difference information. The empirical outcomes firmly confirm the superiority of our streamlined network over nine cutting-edge approaches on the LEVIR-CD, WHU-CD, and SYSU-CD benchmarks, excelling both in terms of accuracy and efficiency. Code and pretrained models are accessible at <https://github.com/SallyRonionGit/DMFANet>.

Index Terms—Building change detection (BCD), difference feature enhancement, multiscale context aggregation, remote sensing (RS).

I. INTRODUCTION

AS ONE of the most typical objects in a city, the distribution of buildings directly reflects the impact of

human activities on the Earth's environment. In the remote sensing (RS) community, building change detection (BCD) aims to identify building changes occurring within a particular geographical area by comparing the images obtained at different observation dates. It has been pervasively utilized across domains like land regulation enforcement [1], building damage assessment [2], and environmental monitoring [3]. With the rapid development of RS technology, very high-resolution (VHR) optical images have become an important data source for BCD, which provides a unique opportunity to monitor dynamic changes of land surface. These images usually not only contain abundant texture and morphological information, but also bring more challenges for BCD tasks, mainly in the following two aspects.

- 1) *Interclass Confusion and Domain Shifts*: The high spectral similarity between buildings and unrelated urban elements (e.g., roads and vehicles) frequently induces misclassification. This issue is exacerbated by inherent limitations in satellite imaging systems. On the one hand, the top-down perspective restricts structural feature extraction, particularly for vertical architectural details. On the other hand, temporal inconsistencies in image acquisition (e.g., varying illumination and seasonal changes) create subtle domain shifts in lighting and image quality between multitemporal pairs, further compromising detection accuracy.
- 2) *Scale-Shape Complexity in Urban Contexts*: Urban environments present significant heterogeneity in building morphologies, where extreme variations in structure scales (from small sheds to skyscrapers) and irregular geometric shapes jointly hinder reliable detection.

To overcome the above issues, numerous BCD methods have been proposed, which can be divided into two categories: the traditional approaches and deep learning (DL)-driven techniques. Traditional BCD approaches for monitoring building changes involve direct comparison [4], [5], image transformation [6], [7], and postclassification comparison [8], [9]. Direct comparison methods calculate the pixelwise differences between bitemporal images through image algebra operations such as subtraction, ratio, and quantification. Image transformation-based methods, including principal component analysis [10] and change vector analysis [11], expertly detect changed regions by transforming the acquired images into a carefully crafted feature space for comparison and analysis. Despite these algorithms' proficiency in identifying differences, they are hindered by accuracy concerns primarily attributed to the simple pointwise processing manner, which

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inherently lacks robust noise suppression capabilities. In addition, it is difficult to choose an optimal threshold to determine meaningful changes that have occurred on the land surface, further compromising their performance. For the postclassification comparison methods, the images acquired at different times are first classified independently, and the classification results are then compared to identify “from-to” types of changes. However, the performance of such methods relies heavily on the accuracy of the classification algorithm. Their failure to capture subtle variations in ground objects and cumulative errors further degrades their effectiveness in real-world scenarios.

Recently, deep neural networks, particularly convolutional neural networks (CNNs) [12], [13], [14], [15], have gained widespread adoption in the field of BCD. By autonomously learning useful features from annotated data, DL-based models mitigate manual intervention and enable end-to-end detection, embodying the promising potential for application in BCD tasks. The preliminary CNN-based methods [16], [17] usually adopt an early fusion structure, concatenating bitemporal images along the channel dimension as the input of model to highlight changes. However, this way easily leads to insufficient mining of semantic information from a single temporal phase. Li and Huo [18] utilized a late fusion architecture, leveraging parallel branches with shared weights to separately extract features from individual temporal image. Despite the powerful feature extraction capability, CNNs still lack the ability to model long-range dependencies, failing to focus on the changed regions of interest in scenarios characterized by imbalanced sample distributions. To this end, an increasing number of studies have introduced attention mechanisms [19], [20], [21] and transformers [22], [23], [24] to model global contextual information. Huang et al. [25] introduced the coordinate attention [26] mechanism, which precisely captures spatial differences between bitemporal features through horizontal and vertical encoding, thereby improving the analysis of spatial context. Feng et al. [27] proposed an intertemporal joint-attention module that combines self-attention and cross-attention, integrating individual representations of bitemporal images to narrow the semantic gaps between their features and suppress changes in irrelevant regions. Although the combination of CNN with attention mechanisms and transformers performs well in BCD tasks, they typically require high-performance hardware to handle the substantial computational overhead, making it difficult to effectively balance model complexity and accuracy.

Current DL-based BCD methods usually consist of the following procedures: 1) bitemporal feature extraction; 2) difference feature generation; 3) multiscale difference feature fusion; and 4) classification. Although these methods have achieved good performance, several critical issues within these steps remain inadequately addressed. First, it is insufficient to explore efficient feature interaction mechanisms in the feature extraction stage. Existing approaches predominantly focus on single-level feature extraction while neglecting cross-level information exchange, making it difficult to comprehensively capture key information of ground objects in complex scenarios, which is crucial for accurately identifying building change

areas. Second, most difference feature generation methods rely on simple operations, such as elementwise subtraction or channel concatenation, to process bitemporal features and extract difference information from a single domain. This manner fails to adaptively capture temporal discrepancies across various feature domains, ultimately leading to the loss of critical change-related information. Third, the process of multiscale difference feature fusion is typically performed in a top-down manner. This strategy struggles to effectively suppress feature redundancy as well as to bridge the semantic gaps between adjacent layers. However, the redundant difference features reduce the contrast between areas with subtle changes and unchanged areas, while the misalignment between layers hinders cross-scale feature learning, resulting in blurred edges with a high rate of missed detection.

To this end, we propose a difference-aware multiscale feature aggregation network (DMFANet) for BCD tasks, designed to not only preserve more details of building changes but also to suppress pseudo-change information. In detail, the acquired bitemporal images are first fed into the MobileNetV2 [28] backbone to extract multilevel features. These features are then refined by the feature modulation module (FMM) to alleviate the issue of domain shift and improve the representation capacity of key information. Subsequently, the cross-domain difference enhancement module (CDEM) is utilized to generate more discriminative difference features through bitemporal feature interaction across channel, spatial, and temporal domains. Afterward, these enhanced difference features are input into the multiscale context aggregation module (MCAM) to fuse and filter out redundant difference information. Finally, a precise binary change map can be produced by a classifier based on convolution with residual structure.

The proposed framework further improves the performance in BCD tasks by overcoming the shortcomings of existing DL-based methods. In the feature extraction stage, the FMM employs deformable convolution (DeConv) with cross-level modulation to resolve inadequate domain alignment and restricted cross-level feature exchange, enabling precise spatial adaptation and enhanced feature integration across levels. In the difference feature generation stage, the CDEM tackles the insufficient expression of difference features within single domain by integrating the difference information from multiple domains, which enhances feature discriminability while reducing interclass confusion. In the difference information fusion stage, the MCAM utilizes a gated fusion with a context-aware aggregation strategy to capture fine-grained change information, effectively bridging semantic gaps across scales.

The main contributions of this work are summarized as follows.

- 1) A lightweight FMM is designed to build the semantically robust and contextually enriched feature representations, which facilitate the matching and differencing of bitemporal features.
- 2) A novel CDEM is developed to efficiently fuse the channel, spatial, and temporal difference information,

thereby producing high-quality difference features for highlighting building changes.

- 3) A transformer-based MCAM is proposed, integrating a duplex gating mechanism to aggregate cross-level contextual difference features while suppressing false detection from redundant difference information.
- 4) The impressive experimental results on three challenging VHR datasets demonstrate that the proposed DMFANet outperforms other state-of-the-art (SOTA) BCD methods with high accuracy and efficacy.

The rest of this article is structured as follows. Section II overviews related works. Section III describes the proposed DMFANet framework in detail. The comprehensive experimental results and analysis are presented in Section IV followed by the discussion of the effectiveness and limitations of the proposed method in Section V. Finally, Section VI concludes this work.

II. RELATED WORK

A. Difference Feature Enhancement Methods

The difference feature enhancement module, a vital component in BCD tasks, achieves strong performance by intrinsically linking its capacity to accurately represent changed regions through the modeling of cross-domain dependencies. Specifically, the channel-domain difference feature emphasizes spectral changes, aiding in the detection of building changes under varying lighting, seasonal, or atmospheric conditions [29], [30]. The spatial-domain difference feature focuses on identifying structural changes in buildings, incorporating more detailed geometric shape information [31], [32]. The temporal-domain difference feature helps to better explore the temporal relationship between bitemporal images for feature interaction [33], [34]. Given this, fully modeling the dependencies of difference features across different domains facilitates the accurate identification of building changes.

Existing difference enhancement methods can be categorized into two main types: single-branch structure and multibranch structure. For the single-branch structures [35], [36], [37], the acquisition of difference information is fixed within a single domain. The differences between features of bitemporal images are computed through cascaded convolutional blocks or traditional arithmetic ways of direct comparison. Such methods are highly sensitive to noise, making it difficult to capture accurate change information. Multibranch structures [19], [27], [38] allow for the parallel processing of bitemporal images, each potentially focusing on different aspects of the input. These branches can then be combined to enhance the difference representation ability that is beneficial for distinguishing real building changes from false changes. However, the methods mentioned above typically perform cross-domain interaction by aggregating multibranch features after capturing difference features within a single domain. During the process of capturing the domain-specific difference features, these methods still fail to consider other observation angles. To improve the quality of difference feature, the strategy of difference information enhancement from different domains is used for BCD tasks, aiming to further

highlight ground changes. In [25] and [39], the enhanced difference information was obtained by summing the difference features from channel and spatial domains based on attention mechanisms, where the former is acquired through channel concatenation and convolution, while the latter is derived from pixelwise subtraction. However, these methods overlook the modeling of temporal relationship between bitemporal images, thereby impairing the ability to identify changed regions in complex scenarios.

B. Multiscale Difference Feature Fusion Techniques

To address the challenge of balancing global information from deep semantic features with fine-grained details of shallow features (such as edges and vertices), multiscale difference feature aggregation modules have been introduced. In [17], [40], and [41], researchers utilized a straightforward top-down feature pyramid network (FPN) structure to conduct feature fusion, which treats features from different layers equally, neglecting their varying contributions to the final result. Moreover, the excessive upsampling without alleviation of the semantic gaps during the context aggregation stage hinders the recovery of fine-grained building edges.

In light of this, features from different layers are reweighted to produce more typical refined ones [25], [42], [43]. However, the reweighting strategies focus solely on the fusion ratio between features at different levels, without considering the significant amount of redundant elements contained within these features. As a result, the network's attention remains primarily on easily detectable building changes, while small-scale changes and those at building edges, which are prone to confusion with roads, are often incorrectly classified as unchanged areas. At present, the transitional decoder structures have become popular, which facilitate the progressive refinement of contextual features, resulting in a more comprehensive and enriched difference information representation. Li et al. [44] proposed a temporal feature interaction and guided refinement (TFI-GR) network, which aggregates temporal difference information from multiscale features to generate a change map in a coarse-to-fine fusion process. Although TFI-GR effectively integrates difference features across layers by pooling them into a unified scale, this process lacks sufficient guidance to bridge the significant semantic gaps between different layers. Furthermore, this approach necessitates repeated module stacking to achieve a satisfactory detection result, which substantially increases the complexity of the model. Consequently, there is an urgent need to explore an efficient multiscale difference feature aggregation mechanism that seamlessly integrates shallow detailed features with deep semantic information, while suppressing the accumulation of redundant features.

III. METHODOLOGY

In this section, the overall encoder-decoder framework of DMFANet is described first. Then, the FMM, CDEM, and MCAM are presented in detail. Finally, the loss function designed for mitigating class imbalance is introduced.

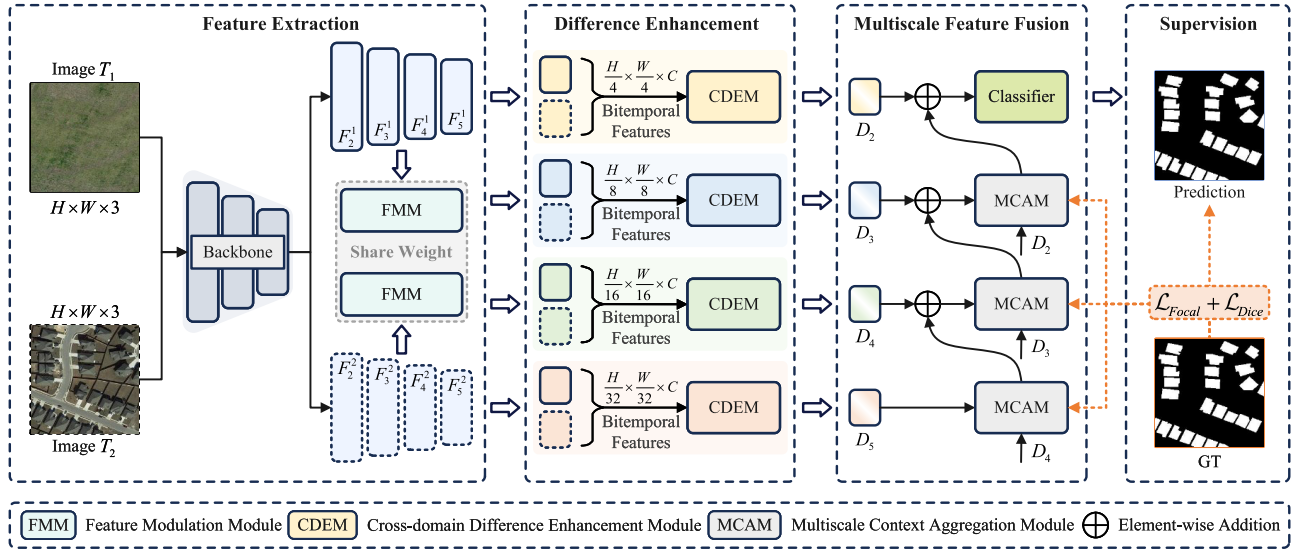


Fig. 1. Overview of the proposed DMFANet. Bitemporal features are first extracted using MobileNetV2 and refined by the FMM through channel-spatial modulation to enhance feature representation ability. The CDEM then models channel, spatial, and temporal dependencies to generate more discriminative difference features for accurate building change localization. Next, the MCAM aggregates multiscale difference features while filtering redundant difference information, ensuring internal consistency and edge details of changed areas. Finally, a convolutional classifier outputs the final binary change map.

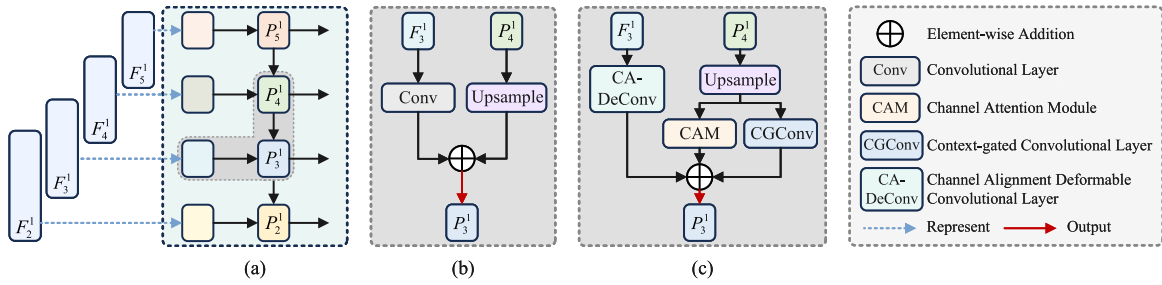


Fig. 2. Scheme of feature modulation process of bitemporal features. (a) Feature transmission within the standard FPN. (b) Cross-level feature interaction in FPN. (c) Cross-level feature interaction in FMM.

A. Architecture Overview

The overall network architecture of the DMFANet framework, based on an encoder-decoder structure, is illustrated in Fig. 1. In the feature extraction stage, bitemporal features are extracted using a lightweight pretrained MobileNetV2 backbone, yielding multilevel features with spatial resolutions of 64×64 , 32×32 , 16×16 , and 8×8 . Subsequently, the FMM standardizes the channel number of each layer to 128 via channel alignment deformable convolutional (CA-DeConv) and modulates these bitemporal features in a bottom-up manner. The CDEM then generates highly discriminative difference features by effectively integrating cross-domain difference information. Following that, the MCAM employs a dynamic gating transformer (DGT) to fuse multiscale difference features and integrates a content-aware aggregation (CAA) module to recalibrate the difference features of adjacent layers, thereby mitigating semantic gaps. Finally, these fused difference features are processed via a convolutional classifier to generate a high-contrast binary change map, reflecting real building changes.

B. Feature Modulation Module

Existing approaches for BCD tasks generally begin with a deep backbone model, composed of multiple processing layers

aimed at learning multilevel feature representations. Features at various levels encode distinct contextual information, which is essential for tackling the inherent challenges in BCD tasks. However, it is of great importance to consider the interaction between cross-level features, which enables the seamless propagation and exchange of information across layers, facilitating the incorporation of complementary information from different levels. This promotes comprehensive matching and differencing between bitemporal features. Inspired by the data flow of the well-known FPN, as depicted in Fig. 2, the FMM refines bitemporal features after feature extraction by utilizing the lateral connections and the bottom-up pathway.

To balance computational efficiency and representation capacity while enabling effective multilevel feature interaction, the FMM unifies the channel numbers of bitemporal features across different levels to 128. Unlike FPN, which relies solely on 1×1 convolution to achieve this, the FMM utilizes the CA-DeConv layer, integrating DeConv [45] and multiscale pyramid convolution (PyConv) [46] to dynamically adjust the number of channels. In contrast to standard convolution, DeConv introduces learned offsets that enable convolution kernels to adaptively adjust sampling locations based on the input features. This mechanism captures more complex geometric shapes and irregular features, thereby addressing

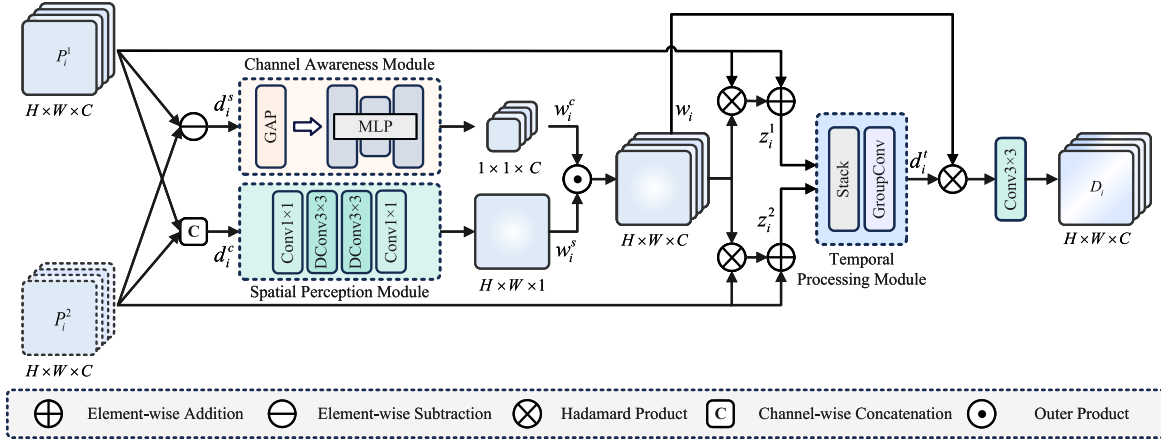


Fig. 3. Scheme of CDEM.

the domain shift problem in bitemporal features. Furthermore, to enrich contextual information, PyConv is incorporated into the learning process of DeConv's offsets, facilitating the extraction of multiscale offsets and expanding the receptive field. Note that PyConv groups features along channels and applies convolution with different kernel sizes for each group, without adding additional computational costs.

After unifying the dimension of features, FPN treats the features from different levels equally and aggregates them in a top-down manner through elementwise addition, enabling cross-level feature interaction. However, this approach consistently fuses adjacent-level features in an equal-proportion manner, disregarding the semantic gaps between features at different levels. Drawn from [47], the performance for small object detection initially increases and then decreases as the fusion ratio of deep and shallow features increases. The attention each layer gives to objects of similar scale should be considered alongside the benefits brought by features from other layers through a broader set of training samples. To achieve smoother multilevel feature interaction, the FMM leverages channel and spatial attention to control the fusion of adjacent features. The channel modulation factor recalibrates the weights along the channel dimension, emphasizing the responses of building-related semantic features. Meanwhile, the spatial modulation factor strategically transfers additional contextually enriched information to the recalibrated features across all spatial locations. To balance model accuracy and computational cost, the FMM employs the lightweight channel attention mechanism (CAM) [48] and context-gated convolution (CGConv) [49] for channel and spatial modulations. By doing so, refined feature representations can be obtained by fusing multilevel features at different scales, which is conducting in capturing fine-grained change information.

C. Cross-Domain Difference Enhancement Module

An exhaustive review of previous works reveals that relying solely on arithmetic calculations and cascaded convolutional blocks for difference feature enhancement is inherently prone to noise interference, and the oversight of modeling cross-domain difference feature dependencies results in missed

detection. Toward this end, the CDEM is designed to dynamically integrate the advantages of difference features across domains by fully merging the difference information from the channel, spatial, and temporal domains.

The CDEM employs a progressive difference generation architecture, comprising three core components: the channel awareness module, the spatial perception module, and the temporal processing module, as shown in Fig. 3. The channel awareness module utilizes elementwise subtraction and absolute value operations to generate spatial difference features, effectively capturing spatial changes. However, its limitation in modeling cross-channel interactions restricts the extraction of discriminative spectral difference information. To overcome this, a squeeze-and-excitation mechanism is incorporated, enabling the module to simultaneously explore both spatial and channel difference information. Through global average pooling (GAP) and multilayer perceptron (MLP), spatial difference features are transformed into a channelwise mask vector, highlighting the channels that are beneficial for BCD tasks. Similarly, for the channel difference features obtained by channel concatenation and convolution, the CDEM employs cascaded dilated convolutions to expand the receptive field and infuse richer spatial difference information. Subsequently, the channel difference features are converted into a spatial mask to assist in the precise localization of building change regions.

In the above process, the channel and spatial masks interact by leveraging the difference features from each other's domain to explore the difference information within their respective domains, implicitly modeling the dependence between the channel and spatial domains. Subsequently, a 3-D weighted mask is explicitly constructed by applying the outer product of both masks. Such operation in the forward propagation phase facilitates the network in capturing critical variation patterns across channel and spatial dimensions, thereby reducing the loss of difference information. Moreover, during the backpropagation stage, it enables the adaptive allocation of gradients to more discriminative feature branches, enhancing the network's learning capability and accelerating convergence. Following that, the enhanced bitemporal features are derived by performing elementwise multiplication of the bitemporal features P_i^1 and P_i^2 with the mask w_i , where $i \in \{2, 3, 4, 5\}$, followed

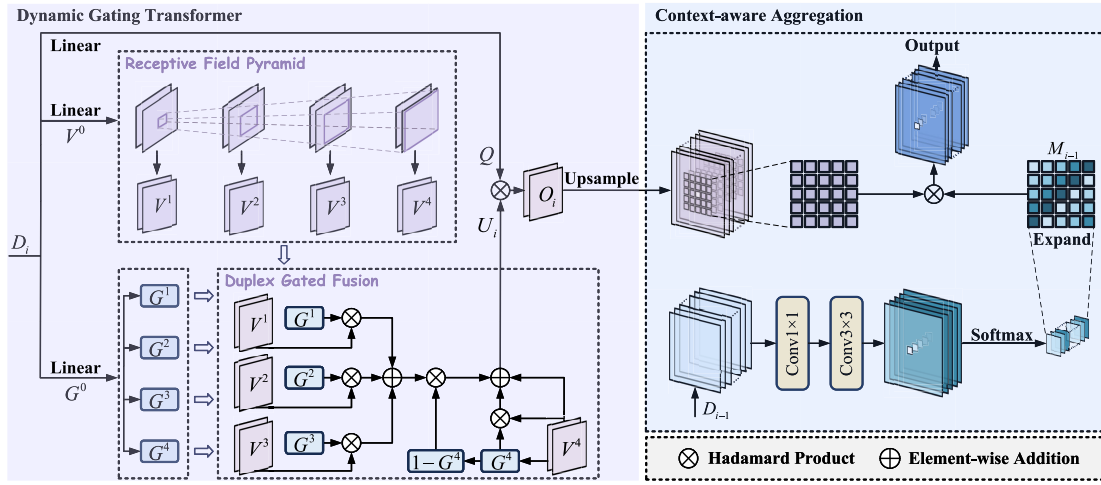


Fig. 4. Scheme of MCAM.

by integration via a residual connection. The complete workflow of this procedure is depicted as follows:

$$d_i^c = \text{Conv}_{1 \times 1}(\text{Conv}_{3 \times 3}(\text{Concat}(P_i^1, P_i^2))) \quad (1)$$

$$d_i^s = |P_i^1 - P_i^2| \quad (2)$$

$$w_i^c = \sigma(\text{MLP}(\text{GAP}(d_i^s))) \quad (3)$$

$$w_i^s = \text{Conv}_{1 \times 1}(\text{DConv}_{3 \times 3}(\text{DConv}_{3 \times 3}(\text{Conv}_{1 \times 1}(d_i^c)))) \quad (4)$$

$$w_i = w_i^s \odot w_i^c \quad (5)$$

$$z_i^1 = P_i^1 \otimes w_i + P_i^1 \quad (6)$$

$$z_i^2 = P_i^2 \otimes w_i + P_i^2 \quad (7)$$

where P_i^1 and P_i^2 represent the refined bitemporal features from the i th layer of the FMM, respectively. d_i^c and d_i^s stand for the channel and spatial difference features in the i th layer. Concat means the channel concatenation, $|\cdot|$ represents the absolute value of subtraction, and σ indicates the Sigmoid activation function. DConv indicates the dilated convolution with the corresponding kernel size followed by the batch normalization layer and rectified linear unit (ReLU) activation function.

In the temporal processing module, the CDEM performs grouped convolution on the initially refined difference features z_i^1 and z_i^2 after interleaving them along the channel dimension, dynamically modeling spatial-temporal dependencies. Finally, the elementwise multiplication of w_i and d_i^t adaptively injects temporal difference information into the channel-spatial dependencies, enabling cross-domain dependence modeling. A subsequent 3×3 convolution is then applied to generate the final difference features D_i . The detailed computation process is given by

$$d_i^t = \text{Group Conv}(\text{Stack}(z_i^1, z_i^2)) \quad (8)$$

$$D_i = \text{Conv}_{3 \times 3}(w_i \otimes d_i^t) \quad (9)$$

where Stack denotes the channelwise interleaved concatenation. Grop Conv signifies the group convolution with the kernel size of 3×3 followed by the instance normalization layer and parametric ReLU (PReLU) activation function.

D. Multiscale Context Aggregation Module

In the decoding stage, effectively leveraging multiscale difference features across layers and suppressing redundant difference features are crucial for ensuring satisfactory BCD performance. On the one hand, the accumulation of redundant difference features increases the contrast between easily detectable building structures and the subtle changes at building edges, which, in turn, diminishes the distinction between subtle change areas and unchanged regions, leading to uneven building edges in the detection results. On the other hand, the semantic gaps between difference features from adjacent layers hinder the learning process of multiscale difference features, making it difficult to detect building changes with different scales.

To address these issues, the MCAM is designed, consisting of two modules: DGT and CAA. The former aims to fuse multiscale difference features and suppress redundant difference information, while the latter is to recalibrate the deep features by leveraging the spatial context derived from shallow features of adjacent layers, thereby effectively bridging the semantic gaps between them. The entire decoder aggregates the difference features layer by layer in the direction of increasing spatial resolution. Finally, the fused features, enriched with contextual information, are passed into a classifier to generate the final binary change map.

From Fig. 4, it is clear that the DGT consists of a receptive field pyramid and a duplex-based gating fusion mechanism. Inspired by [50], we transform the original complex self-attention mechanism in transformer into a modulation process, where difference features are used to query. Specifically, given the difference features D_i at the i th layer, the query Q , the vanilla features V^0 , and the gating mask G^0 can be obtained by

$$V^0 = f_v(D_i), \quad Q = f_q(D_i), \quad G^0 = f_g(D_i), \quad i = 3, 4, 5 \quad (10)$$

where f_v , f_q , and f_g are the corresponding linear mapping functions. To achieve hierarchical feature representations, we design a receptive field pyramid-based convolutional

structure, consisting of three convolutional blocks and a GAP block. Each convolutional block incorporates a depthwise convolution (DWConv) with an adaptive kernel size, followed by the Gaussian error linear unit (GeLU) activation function, to extract local difference information. Meanwhile, the GAP block captures global difference information. The detailed process is given by

$$V^k = \text{GeLU}(\text{DWConv}(V^{k-1})), \quad k = 1, 2, 3 \quad (11)$$

$$V^4 = \text{GAP}(V^3). \quad (12)$$

To further fuse these difference features while suppressing the accumulation of redundant difference information, a duplex-based gating fusion mechanism is employed. For the difference features obtained from different receptive fields, a corresponding gating mask performs feature selection, filtering out irrelevant information and focusing on the changed regions. The duplex gated difference feature fusion process is outlined as follows:

$$U_i = (1 + G^4) \otimes V^4 + (1 - G^4) \otimes \sum_{k=1}^3 G^k \otimes V^k \quad (13)$$

where U_i denotes the fused difference features of the i th layer. V^k represents the difference features extracted from the k th receptive field, and G^k is the corresponding gate mask, which is ranged from 0 to 1 by using the Sigmoid activation function.

In the fusion process, the difference features of the first three layers (i.e., V^1 , V^2 , and V^3) act as the sender, while the difference features of the last layer (i.e., V^4) servers as the receiver. When valuable difference information from a specific sender is missing at the receiver (i.e., the sender's pixel values are high while the receiver's are low), the gating unit adjusts accordingly: it increases the gating value of the sender and decreases that of the receiver, enabling the missing difference information to be transmitted. Once the receiver obtains the previously missing information, its gating value increases while the senders' gating values decrease, limiting further transmission and reception to prevent redundant information accumulation. This gating mechanism maintains the hierarchical importance of distinct differences, ensuring the integrity of each component.

After fusing these difference features with different receptive fields, the U_i is transformed through a 1×1 convolutional layer into a modulation mask for the query Q , which is then elementwise multiplied to obtain the enhanced difference features. The computation process is described as follows:

$$O_i = Q \otimes \text{Conv}_{1 \times 1}(U_i) \quad (14)$$

where O_i is the output result of DGT at the i th layer.

To alleviate the semantic gaps between difference features from different layers, a CAA module is developed. It first performs preliminary feature extraction on the upper features D_{i-1} through two convolutional layers with kernel sizes of 1×1 and 3×3 . Then, a softmax function is used to normalize the features along the channel dimension, and each channel vector of center pixel is reshaped into a local attention mask M_{i-1} , which is multiplied O_i after upsampling to obtain the aggregated difference features at the i th layer. For example,

in the bottom MCAM, which takes the difference features D_4 and D_5 as input, the generated difference features can be obtained by

$$M_4 = \text{softmax}(\text{Conv}_{3 \times 3}(\text{Conv}_{1 \times 1}(D_4))) \quad (15)$$

$$\text{DF}_4 = M_4 \otimes \text{Upsample}(O_5) \quad (16)$$

where DF_4 denotes the output difference features of the bottom MCAM. Through the progressive processing procedure, high-quality difference features are extracted and then fed into a classifier to predict actual building changes.

E. Loss Function

For the task of BCD in the RS community, the issue of class imbalance often arises as the unchanged areas significantly outnumber the changed ones. To solve this problem and guide the model to better handle varying scale changes of buildings, a hybrid loss function that incorporates both Focal loss [51] and dice coefficient (Dice) loss [52] is employed. Specifically, the Focal loss and Dice loss can be expressed as

$$\mathcal{L}_{\text{Focal}}(\hat{y}, y) = -\alpha(1 - \hat{y})^\gamma \log(\hat{y}) \quad (17)$$

$$\mathcal{L}_{\text{Dice}}(\hat{y}, y) = 1 - \frac{2y\hat{y} + 1}{y + \hat{y} + 1} \quad (18)$$

where y and $\hat{y}$ are the ground truth (GT) and predicted change map, respectively. The value of pixel in y is 0 or 1, while the value of pixel in $\hat{y}$ ranges from 0 to 1, indicating the probability of being predicted as the change class. α represents the weighting factor, and γ denotes the tunable focusing parameter. The overall loss function is calculated as follows:

$$\mathcal{L}_{\text{Total}} = \sum_{i=2}^5 \lambda \mathcal{L}_{\text{Focal}}(\hat{y}_i, y) + \mathcal{L}_{\text{Dice}}(\hat{y}_i, y) \quad (19)$$

where $\hat{y}_i$ corresponds to the predicted change map of the decoder at the i th stage, and λ is used to balance the two loss terms.

IV. EXPERIMENTAL RESULTS AND ANALYSIS

A. Datasets

In our experiments, three public benchmark BCD datasets were used, with their comprehensive details outlined as follows.

- 1) *LEVIR-CD [20] Dataset*: This dataset specifically tailored for BCD, comprises 637 pairs of bitemporal RS images, and each image is with the size of 1024×1024 and a spatial resolution of 0.5 m. These images were sourced from 20 distinct regions in TX, USA, utilizing the Google Earth platform. Adhering to the official dataset split configuration, the images were processed by cropping them into nonoverlapping patches of 256×256 pixels, resulting in a distribution of 7120, 1024, and 2048 image pairs for training, validation, and testing, respectively.
- 2) *WHU-CD [53] Dataset*: This dataset is comprised of a pair of optical images acquired from 2012 and 2016 with the size of 32507×15354 and share the same spatial

resolution of 0.2 m. Following the rule outlined in [35], this expansive image was segmented into nonoverlapping patches with pixel dimensions of 256×256 . Subsequently, through random partition, these patches were divided into training, validation, and testing sets, resulting in 4536, 504, and 2760 pairs, respectively.

- 3) *SYSU-CD [5] Dataset*: This dataset consists of 20 000 pairs of 256×256 pixels bitemporal RS images, each with a 0.5-m spatial resolution, captured from Hong Kong between 2007 and 2014. It showcases diverse and complex change scenarios like road expansions, new buildings, vegetation changes, and suburban growth. The official 6:2:2 split divides the dataset into 12 000, 4000, and 4000 pairs for training, validation, and testing, respectively.

B. Implementation Details and Evaluation Metrics

1) *Implementation Details*: In our experiments, the proposed DMFANet was implemented by the PyTorch platform and trained for 200 epochs on an NVIDIA RTX 4090 GPU with 24 GB of memory, employing a batch size of 16. A cosine annealing learning rate schedule was implemented throughout the training process, with an initial learning rate of 0.0005 and a 10% warm-up phase at the beginning of training. In the total loss function, we used the recommended value of 0.25 for the parameter α , where γ and λ are set to 4 and 0.5, respectively. AdamW with the weight decay of 0.0005 was chosen as the optimizer. For the sake of fairness, data augmentation strategies, including flipping, cropping, random temporal exchanging, and Gaussian blurring, were consistently applied on the proposed framework and other competitors in our experiments.

2) *Evaluation Metrics*: To quantitatively evaluate the performance of BCD methods, five prevalent evaluation metrics including precision (P), recall (R), overall accuracy (OA), $F1$ -score ($F1$), and intersection over union (IoU) were used, which can provide a comprehensive assessment of the accuracy, completeness, and overall effectiveness of different BCD methods. The above evaluation indicators are given as follows:

$$P = TP / (TP + FP) \quad (20)$$

$$R = TP / (TP + FN) \quad (21)$$

$$OA = (TP + TN) / (TP + TN + FP + FN) \quad (22)$$

$$F1 = 2 \times P \times R / (P + R) \quad (23)$$

$$IoU = TP / (TP + FP + FN) \quad (24)$$

where TP, TN, FP, and FN denote the quantities of true positives, true negatives, false positives, and false negatives, respectively. The values of these indicators range from 0 to 1, with elevated values signifying superior performance.

C. Comparative Methods

To verify the effectiveness of the proposed DMFANet, nine SOTA BCD methods are selected for comparison in our experiments. It is essential to emphasize that all methods were rigorously reevaluated under identical experimental conditions, strictly following the official protocols established by the respective authors. These methods are summarized as follows.

- 1) *FC-EF [17]*: This method utilizes an early fusion architecture to integrate bitemporal images and employs a fully convolutional network for monitoring building changes.
- 2) *SNUNet [21]*: Leveraging a multiscale feature extraction mechanism, SNUNet effectively captures spatial and temporal changes, enhancing the model's focus on relevant changes while suppressing noise interference.
- 3) *IFNet [54]*: It introduces a deeply supervised network tailored for BCD, employing a multiscale feature fusion strategy to reconstruct an accurate change map.
- 4) *HEANet [55]*: By incorporating high-frequency cues through the isotropic Sobel operator, this approach can detect fine-grained changes, through capturing more precise and detailed edge information.
- 5) *BIT [56]*: This approach compresses high-level semantic information into tokens and introduces both self-attention and cross-attention mechanisms to capture long-range dependencies in the spatial-temporal domain.
- 6) *ChangeFormer [22]*: Based on a Siamese network architecture, ChangeFormer combines a hierarchical transformer encoder with a lightweight MLP decoder to effectively render multiscale long-range details.
- 7) *USSFC [37]*: This method utilizes multiscale decoupled convolution to extract representative features of ground objects with different shapes and effectively captures global contextual information to refine features by modeling spatial-spectral relationships of changed regions.
- 8) *DMINet [27]*: This method combines CNNs with transformers to mitigate semantic gaps between bitemporal images and detect changes through a multibranch difference feature enhancement framework.
- 9) *SEIFNet [25]*: It employs coordinate attention into multibranch structure based on subtraction and cascaded convolution to enhance spatial and temporal difference features. Furthermore, a progressive decoder is adopted to retain boundary details with a complete internal structure.

D. Performance Comparison and Results Analysis

1) *Qualitative Assessment*: To validate the effectiveness of the proposed DMFANet, detailed visual BCD results across different datasets with various scenarios and scales are illustrated in Figs. 5–7. The visualizations employ a color-coding scheme to distinguish the veracity of the outcomes, categorizing them into TP (signified by white), TN (rendered in black), FP (highlighted in red), and FN (denoted by green).

In BCD tasks, structural buildings are frequently occluded by vegetation shadows or misidentified due to spectral similarities between asphalt surfaces, rooftops, and building facades, which consequently leads to increased false alarm rates. As illustrated in Figs. 5(a) and 6(d), compared to other methods, DMFANet is the only method that exhibits exceptional discriminability, even under occlusion and illumination interference. Moreover, as shown in Fig. 7(e), DMFANet exhibits a significant advantage in maintaining the internal integrity and well-defined boundaries of buildings, fully substantiating its robustness against various interference factors.

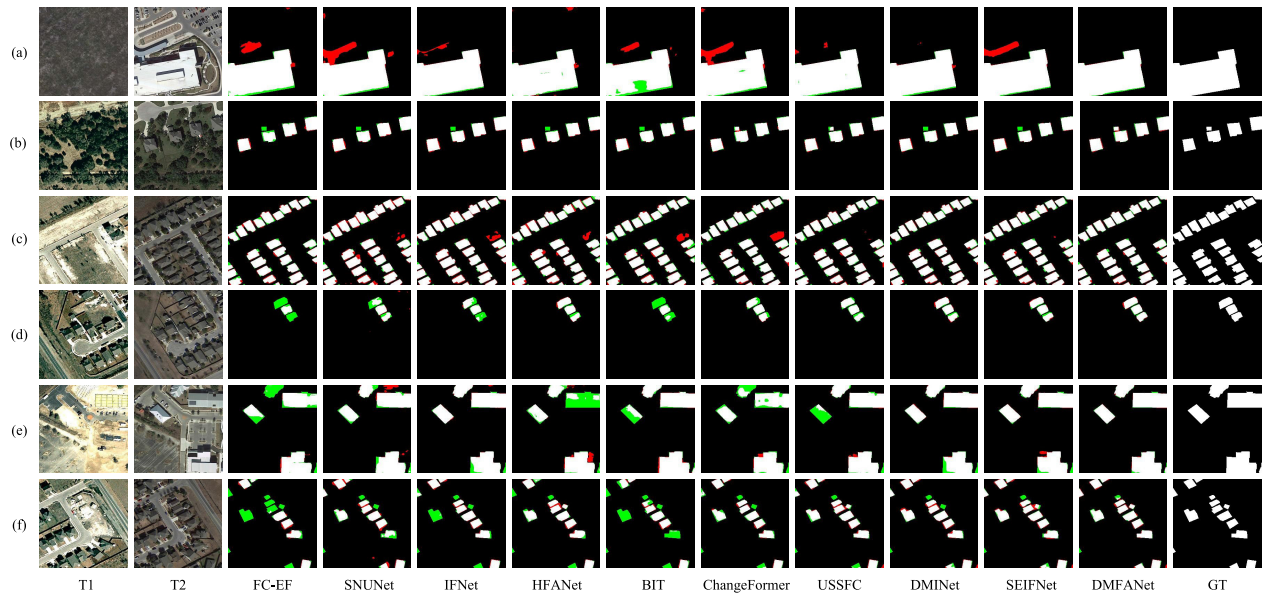


Fig. 5. Visual results of different BCD methods on the LEVIR-CD dataset. (a)–(f) Six representative samples, where white and black represent the changed and unchanged areas, while red and green indicate the false detection pixels and missed detection pixels, respectively.

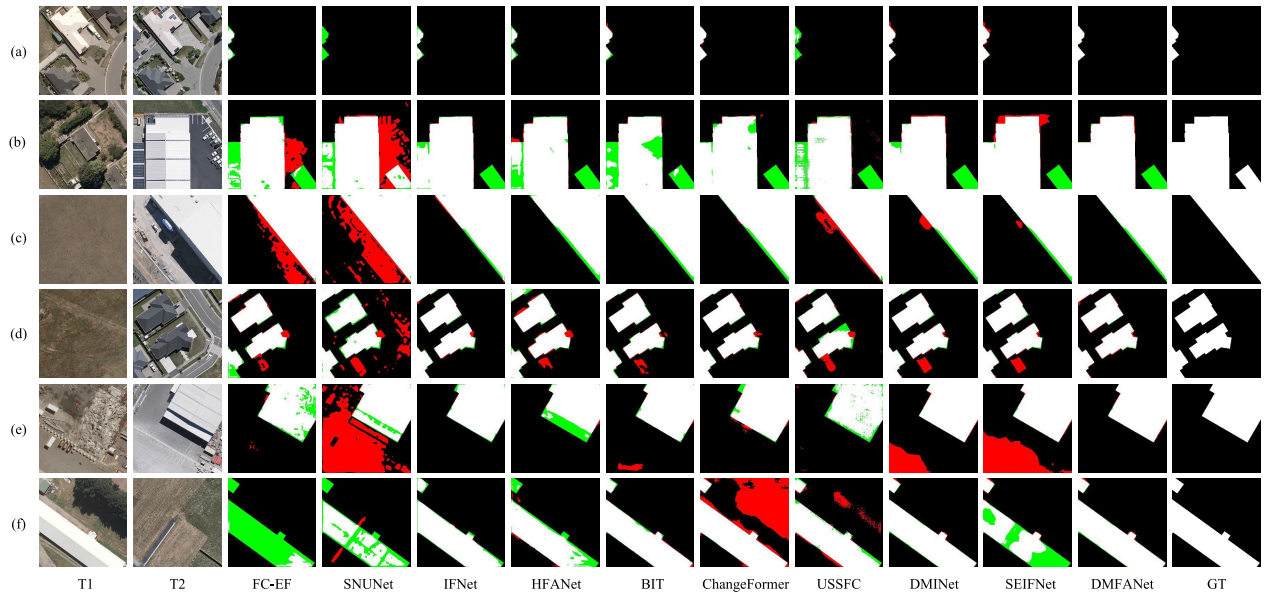


Fig. 6. Visual results of different BCD methods on the WHU-CD dataset. (a)–(f) Six representative samples, where white and black represent the changed and unchanged areas, while red and green indicate the false detection pixels and missed detection pixels, respectively.

A critical challenge in current BCD tasks lies in resolving the conflict between structural fidelity preservation of large-scale building changes and discriminative feature retention of small-scale alterations under unified frameworks. For instance, as shown in Fig. 7(b), due to the limited receptive field inherent in convolutional operations, large buildings are prone to being missed by other methods, resulting in blurred edges. However, DMFANet successfully achieves detection results with well-preserved internal structures and clear boundary details through its superior capability in capturing and combining multiscale difference features. Furthermore, it effectively identifies small-scale building changes in Fig. 5(d), demonstrating exceptional discriminative

capability in detecting subtle changes. The visualization results over three datasets demonstrate that our method can accurately identify changes with different scales and shapes of buildings, achieving superior CD performance compared to other methods.

2) *Quantitative Evaluation*: The comparative assessments of various algorithms across the LEVIR-CD, WHU-CD, and SYSU-CD datasets are summarized in Table I. The rigorous evaluation framework encompasses pivotal metrics including P , R , OA , $F1$, and IoU , comprehensively evaluating the performance of different algorithms.

On the LEVIR-CD dataset, DMFANet outperforms the latest SEIFNet and DMINet, achieving a significant increase in

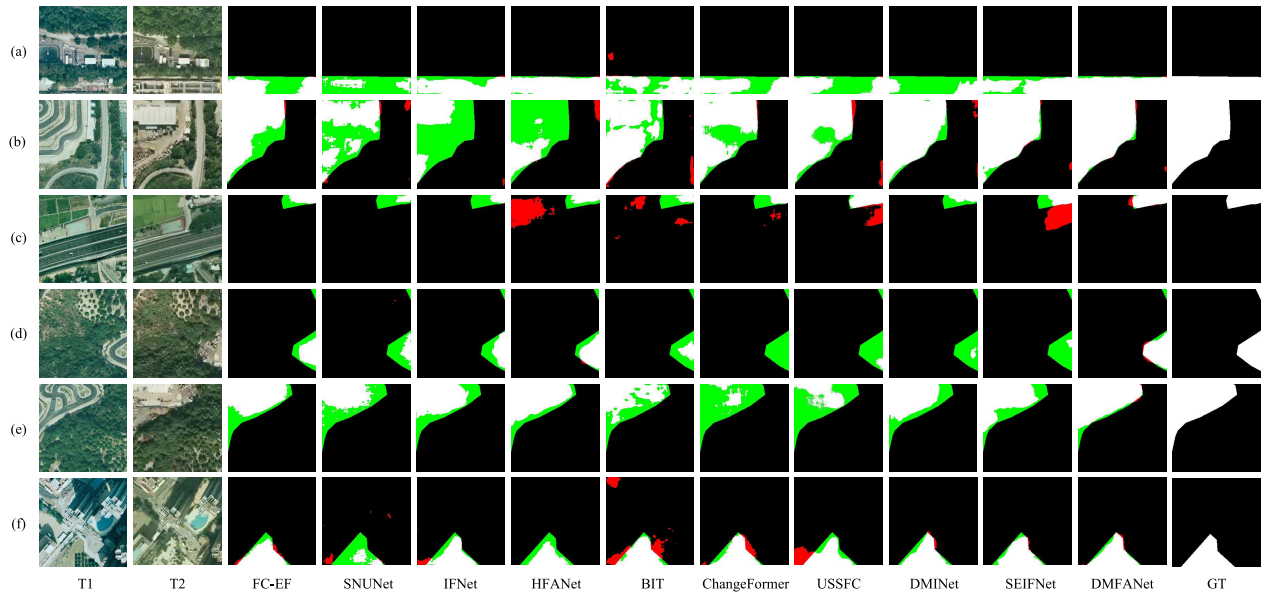


Fig. 7. Visual results of different BCD methods on the SYSU-CD dataset. (a)–(f) Six representative samples, where white and black represent the changed and unchanged areas, while red and green indicate the false detection pixels and missed detection pixels, respectively.

$F1$ and IoU , reaching 92.09% and 85.33%, respectively. This is due to that it establishes semantically robust and contextually enriched hierarchical feature representations during the bitemporal feature extraction process, making it better at locating building change regions that are often overlooked by other methods. Furthermore, DMFANet is 1.3% and 2.19% higher than the second-ranked IFNet in terms of $F1$ and IoU indicators, further verifying its effectiveness in building changes with different scales. On the WHU-CD dataset, DMFANet achieves 99.52%, 93.10%, and 87.08% in OA , $F1$, and IoU , which are 0.3%, 3.5%, and 5.91% higher than the second-ranked IFNet, respectively. Although our method does not obtain the best results in terms of P and R , the overall performance is superior to other competitors. This may be because CDEM can fully explore the relationship between difference information from different domains and MCAM can fully fuse the enhanced multilevel difference information, thereby producing more discriminative feature representations. On the SYSU-CD dataset, DMFANet achieves 80.72%, 92.33%, 83.23%, and 71.27% in terms of P , OA , $F1$, and IoU , which are 2.47%, 1.04%, 2.32%, and 3.33% higher than the second-best SEIFNet, respectively. This demonstrates that the proposed approach exhibits improved effectiveness in simultaneously reducing both commission and omission errors. In conclusion, the quantitative assessments conducted conclusively validate the effectiveness and preeminence of DMFANet. The notable improvements across all evaluated metrics and datasets provide compelling evidence of DMFANet's strengths, positioning it as a formidable tool for tackling complex BCD tasks in RS imagery.

3) *Complexity Analysis*: To comprehensively evaluate the complexity of the model, the number floating point operations (FLOPs) and parameters (Params) are selected as the assessment indicators, which are used to measure the computational workload and the size of parameters in the model, respectively.

Table I lists the concrete numerical metrics of different methods. Obviously, the proposed DMFANet achieves over 2% performance improvement on the LEVIR-CD dataset while utilizing only about one-sixth of the parameters compared to the high-complexity ChangeFormer and HFANet models, demonstrating its exceptional efficiency. Compared to DMINet, an increase of less than 1M Params in DMFANet leads to $F1$ and IoU improvements of 7.65% and 4.56% on the WHU-CD dataset, respectively. When compared to the lightweight BIT method, despite having twice the number of parameters, the proposed approach achieves an approximately 7% improvement in IoU on the SYSU-CD dataset. Moreover, in terms of FLOPs, the computational cost remains within an acceptable range, aligning with the average level among all methods. These results highlight the strong balance between performance and efficiency achieved by DMFANet, making it a competitive choice for BCD tasks.

V. DISCUSSION

A. Analysis of λ and γ

In the proposed framework, parameters λ and γ in the total loss function have an important influence on the performance of the model. To this end, we conducted an analysis by systematically adjusting both parameters on the WHU-CD dataset. As shown in Table II, the weight parameter λ was set to 0.5, 1, and 2, while the tunable focusing parameter γ was configured as 2, 4, and 6. When γ is fixed, the CD performance of DMFANet slowly decreases as λ increases. When λ is fixed, the CD performance of the model first increases and then decreases as γ increases. From the experimental results in Table II, it can be observed that DMFANet achieves the best performance when the weight parameter λ is set to 0.5 and the focusing parameter γ is set to 4. Therefore, in this study, we set λ and γ to 0.5 and 4 for all subsequent experiments.

TABLE I

QUANTITATIVE COMPARISONS ON THREE DATASETS. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD. ALL INDICATORS ARE DESCRIBED AS %

Method		FC-EF [17]	SNUNet [21]	IFNet [54]	HFANet [55]	BIT [56]	ChangeFormer [22]	USSFC [37]	DMINet [27]	SEIFNet [25]	DMFANet (Ours)
FLOPs(G)		7.15	109.67	164.52	258.24	21.27	405.58	9.72	29.10	16.74	38.48
Params(M)		1.35	12.03	50.44	61.99	3.50	41.03	1.52	6.24	27.91	7.18
LEVIR-CD (%)	P	89.36	85.14	91.55	88.62	91.46	91.95	91.69	91.73	91.77	93.27
	R	81.55	87.92	90.05	86.03	81.76	87.15	89.66	89.51	89.12	90.93
	OA	98.57	98.60	99.07	98.73	98.68	98.96	99.06	99.06	99.04	99.20
	F1	85.28	86.51	90.79	87.31	86.34	89.49	90.66	90.61	90.43	92.09
	IoU	74.33	76.23	83.14	77.47	75.96	80.98	82.92	82.83	82.53	85.33
WHU-CD (%)	P	86.17	53.23	95.12	90.45	85.92	90.86	85.79	85.40	86.23	94.45
	R	76.30	80.18	84.69	82.44	86.23	84.19	77.82	91.92	90.96	91.78
	OA	98.57	96.42	99.22	98.96	98.89	99.04	98.61	99.06	99.07	99.52
	F1	80.94	63.98	89.60	86.26	86.07	87.40	81.61	88.54	88.53	93.10
	IoU	67.98	47.04	81.17	75.84	75.55	77.62	68.93	79.43	79.42	87.08
SYSU-CD (%)	P	80.50	78.96	85.02	81.99	78.78	91.77	82.73	86.66	83.76	85.90
	R	73.73	70.92	72.09	74.38	77.42	72.59	77.60	75.79	78.25	80.72
	OA	89.59	88.68	90.42	90.10	89.76	89.72	90.90	91.54	91.29	92.33
	F1	76.97	74.72	78.02	77.80	78.10	76.91	80.08	80.86	80.91	83.23
	IoU	62.56	59.64	63.97	63.93	64.06	62.48	66.78	67.87	67.94	71.27

TABLE II

QUANTITATIVE COMPARISONS OF DIFFERENT PARAMETER COMBINATIONS ON THE WHU-CD DATASET

λ	γ	P	R	OA	F1	IoU
0.5	2	95.05	87.45	99.39	91.09	83.63
1	2	95.19	87.22	99.39	91.03	83.54
2	2	94.35	86.59	99.34	90.30	82.32
0.5	4	94.45	91.78	99.52	93.10	87.08
1	4	94.21	91.64	99.50	92.90	86.75
2	4	95.25	89.21	99.46	92.13	85.41
0.5	6	95.83	90.30	99.52	92.98	86.88
1	6	94.91	89.98	99.47	92.38	85.88
2	6	96.40	88.16	99.46	92.09	85.35

TABLE III

 z -VALUE OF McNEMAR'S TEST BETWEEN DMFANET AND OTHER METHODS

Method	Dataset		
	LEVIR-CD	WHU-CD	SYSU-CD
FC-EF [17]	22.91	2.91	68.10
SNUNet [21]	130.51	278.07	288.91
IFNet [54]	24.17	57.59	195.40
HFANet [55]	60.46	7.77	42.69
BIT [56]	2.44	60.05	339.13
ChangeFormer [22]	30.93	4.13	127.97
USSFC [37]	39.66	67.13	34.14
DMINet [27]	18.73	18.79	239.65
SEIFNet [25]	16.60	5.27	62.80

B. Significant Statistical Analysis

To further validate the statistical significance of the experimental outcomes, McNemar's test [57] is utilized to examine the accuracy enhancement achieved by DMFANet in comparison with another baseline method. The z -statistic is formulated as

$$z = \frac{|N_{12} - N_{21}|}{\sqrt{N_{12} + N_{21}}} \quad (25)$$

where N_{12} denotes the number of samples correctly classified by method 1 but misclassified by method 2, whereas N_{21} represents the number of samples correctly classified by method 2 but misclassified by method 1. When $z > 1.96$, the null hypothesis (i.e., there is no significant performance difference between the two methods) can be rejected with 95% confidence level. Table III presents the z -values for comparisons between DMFANet and other methods. The results indicate that the accuracy improvement achieved by DMFANet is statistically significant in comparison with other approaches.

C. Ablation Study

1) *Effectiveness of FMM*: The FMM introduces an effective design optimization to the traditional encoder-decoder-based BCD framework, providing a concise yet efficient channel-spatial modulation mechanism for multilevel bitemporal feature interaction during the feature extraction phase. To validate the efficiency of FMM, we compared it with the FPN illustrated in Fig. 2(b), termed w/o FMM. Specifically, FPN aligns the spatial dimensions of adjacent layer features through upsampling, followed by elementwise addition and convolution to achieve feature aggregation. The experimental results presented in Table IV (#02) reveal that the incorporation of FMM leads to substantial performance enhancements on the WHU-CD dataset, demonstrating improvements of 1.6% in $F1$ and 2.75% in IoU compared to the baseline model w/o FMM. The performance improvement stems from the encoder's cross-level interaction mechanism, which strengthens multiscale feature learning while enabling thorough extraction of hierarchical semantic information, ultimately yielding enhanced discriminative representations for effective difference feature analysis.

TABLE IV

QUANTITATIVE COMPARISONS OF THE PROPOSED METHOD WITH DIVERSE SETTINGS ON THREE DATASETS. ALL INDICATORS ARE DESCRIBED AS %

No.	Variants	LEVIR-CD(%)					WHU-CD(%)					SYSU-CD(%)				
		P	R	OA	F1	IoU	P	R	OA	F1	IoU	P	R	OA	F1	IoU
#01	DMFANet	93.27	90.93	99.20	92.09	85.33	94.45	91.78	99.52	93.10	87.08	85.90	80.72	92.33	83.23	71.27
FMM																
#02	w/o FMM	92.21	90.71	99.14	91.46	84.26	95.22	88.02	99.42	91.50	84.33	87.34	78.73	92.29	82.81	70.66
#03	w/o CA-DeConv	92.83	90.96	99.18	91.89	84.99	93.59	88.04	99.36	90.73	83.03	84.67	81.25	92.11	82.92	70.82
CDEM																
#04	Sub	92.93	90.68	99.17	91.79	84.83	96.21	85.89	99.38	90.76	83.08	86.22	79.70	92.21	82.83	70.69
#05	Concat	92.51	91.35	99.18	91.93	85.06	95.22	88.34	99.43	91.65	84.59	84.70	78.65	91.61	81.56	68.87
#06	SC-DEM	92.60	91.10	99.18	91.85	84.92	95.07	86.25	99.35	90.44	82.55	86.74	78.45	92.09	82.39	70.05
MCAM																
#07	w/o MCAM	93.04	90.18	99.16	91.59	84.48	94.28	90.66	99.47	92.43	85.93	88.04	78.09	92.33	82.77	70.60
#08	w/o DGT	93.10	90.63	99.17	91.85	84.97	95.86	89.42	99.49	92.53	86.10	87.41	78.69	92.30	82.82	70.68
#09	w/o CAA	92.58	91.21	99.18	91.89	85.06	94.63	90.41	99.48	92.47	85.99	86.45	79.56	92.31	82.86	70.75

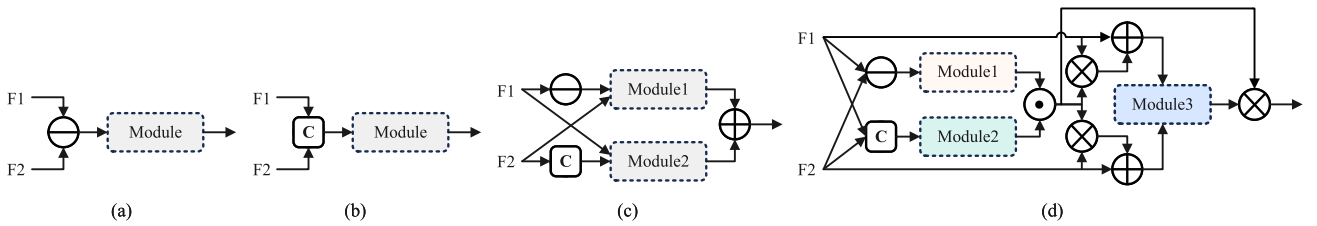


Fig. 8. Simplified schematic of different difference feature enhancement modules, where $F1$ and $F2$ are the features from bitemporal images. (a) Difference-based module (Sub). (b) Connection-based module (Concat). (c) Multibranch module (SC-DEM). (d) Cross-domain module (CDEM).

Furthermore, to validate the impact of CA-DeConv on the model performance, we replaced it with a standard 3×3 convolution operation in FMM, termed w/o CA-DeConv. As shown in Table IV (#03), CA-DeConv demonstrates significant performance gains over w/o CA-DeConv, achieving improvements of 2.37% in $F1$ and 4.05% in IoU on the WHU-CD dataset. This performance enhancement can be attributed to CA-DeConv's dynamic sampling with offsets, which enables superior feature alignment compared to the fixed receptive field of standard convolution, thereby ensuring consistent feature fusion after channel unification.

2) *Effectiveness of CDEM*: The proposed DMFANet integrates CDEM to enhance the representation capability of difference features. To thoroughly validate its effectiveness, we carefully designed comparative experiments, comparing it with the single-branch structure and multibranch structure detailed in Section II. As illustrated in Fig. 8(a) and (b), difference-based feature enhancement methods like ensemble spatial-channel attention fusion (ESCAF) [40] and connection-based approaches such as the cross-temporal fusion module (CTFM) [31] both fall under the category of single-branch difference enhancement structures. In the ESCAF, difference features are enhanced by elementwise subtraction followed by channel and spatial attention module. In the CTFM, after channel concatenation, bitemporal features are refined with a 3×3 convolution to enhance the representation ability of difference features. To simplify the above processes, we exclude optional modules such as attention mechanisms, which are implemented differently across various studies. Instead, we focus solely on the common difference

feature enhancement architectures, direct elementwise subtraction (Sub), and convolution after channel connection (Concat), as an alternative to the CDEM. They are, respectively, constrained to generating difference features through spatial and channel domains.

As shown in Table IV, on the WHU-CD dataset, the CDEM achieves remarkable improvements of 2.34% and 4.00% in $F1$ and IoU compared to the spatial-domain Sub (#04), and the gains of 1.45% and 2.49% compared to the channel-domain Concat (#05), respectively. Similarly, notable enhancements can be observed on the LEVIR-CD and SYSU-CD datasets. These improvements primarily attributed to CDEM's effective modeling of dependence relationships in channel-spatial difference features, which addresses two critical limitations: the absence of interchannel dimension interactions in the Sub approach and the constrained receptive field in the Concat method. Furthermore, to better illustrate the efficacy of modeling the cross-domain difference feature dependencies, we compared CDEM with the spatial-channel difference enhancement module (SC-DEM) [25], a specific implementation of the multibranch structure shown in Fig. 8(c), which aggregates features from spatial-channel domains through superposition but overlooks the temporal difference information. From Table IV (#06), obviously, CDEM enhances the discriminative characteristics of changed regions by modeling the dynamic evolutionary relationships between bitemporal features, thereby achieving optimal performance. The comparative experimental results sufficiently demonstrate that the joint learning of difference features across multiple domains, which compensates for the missing difference information in a single domain by incorporating complementary information

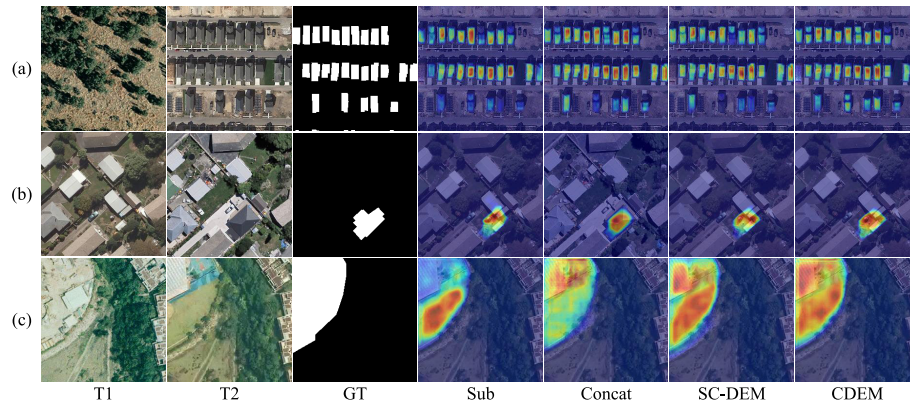


Fig. 9. Visual comparisons of the CDEM with Sub, Concat, and SC-DEM on three datasets. (a) LEVIR-CD. (b) WHU-CD. (c) SYSU-CD.

from other domains, is crucial for improving the BCD performance.

3) *Effectiveness of MCAM*: In the proposed MCAM, multiscale difference features at each layer are effectively extracted and refined through the DGT and CAA modules. To verify its effectiveness, we simplified it into a pure top-down FPN structure (termed w/o MCAM) for comparison. The results in Table IV (#07) show that DMFANet w/o MCAM performs worse than the full DMFANet, highlighting MCAM's ability to adapt to building scale variations and preserve edge details. Additionally, we designed two DMFANet variants: one using MCAM without the DGT module (termed w/o DGT) and the other without the CAA module (termed w/o CAA) to assess each component's effectiveness. As shown in Table IV (#08) and (#09), both variants outperform DMFANet w/o MCAM, with IoU gains of 0.49% and 0.58% on the LEVIR-CD dataset, but perform worse than the full DMFANet. This demonstrates that a single module alone is insufficient for capturing comprehensive difference information. The DGT module effectively captures multireceptive field difference features while suppressing redundant difference information. Meanwhile, the CAA module adaptively recalibrates the pixel distribution of deep features by integrating contextual information from adjacent shallow layers, bridging the semantic gaps between difference features at different scales. By efficiently integrating both modules, MCAM further enhances DMFANet's performance, enabling more accurate detection results.

D. Visualization of CDEM

In Fig. 9, a series of heatmaps generated using Grad-CAM [58] is presented, with the postevent image and heatmaps overlaid together. The heatmaps of the DMFANet illustrate the attention distribution in the last convolutional layer after difference enhancement by CDEM and other comparison algorithms. As shown in Fig. 9(a), on the LEVIR-CD dataset, other difference feature enhancement methods fail to detect subtle changes in densely populated building clusters, whereas CDEM successfully identifies them. Furthermore, upon examining the heatmaps in Fig. 9(b) and (c), it becomes apparent that whether dealing with isolated small target changes or large-scale building changes, CDEM provides a

more comprehensive representation of the internal structure of buildings and delineates clearer boundaries. These observation underscores that modeling cross-domain dependence relationships among difference features enhances the extraction of domain-specific discriminative characteristics, thereby significantly improving the accuracy of building change region identification.

E. Limitations

Although the proposed DMFANet has achieved superior performance on three challenging datasets compared to other DL-based methods, it also has some drawbacks. It requires a large amount of labeled samples for supervised training, whereas the process of obtaining these data is labor-intensive and time-consuming, resulting in the inadaptability of DMFANet in BCD tasks with few labeled data. Furthermore, the proposed model is trained and tested on datasets with the same distribution, and directly transferring it to a new dataset often leads to a sharp decline in performance, especially in the dataset with inconsistent distribution. In addition, the proposed framework employs a lightweight backbone model for feature extraction, which may make the semantic information of bitemporal features slightly insufficient, and more advanced models such as transformer and Mamba can be introduced to further improve the capability of feature representation and enhance its CD performance in complex scenarios.

VI. CONCLUSION

In this study, we propose a novel DMFANet framework incorporating three key components: FMM, CDEM, and MCAM for BCD tasks. During the bitemporal feature extraction phase, the FMM enhances feature representation ability through multilevel feature interactions. Then, the CDEM is used to obtain enhanced difference information by modeling interdependencies among cross-domain difference features and effectively integrating their complementary advantages. In addition, to relieve the semantic gaps across different levels and aggregate multiscale difference features, the MCAM is integrated to improve the model's ability to capture building changes at different scales. Extensive experiments conducted

on three benchmark datasets demonstrate that the proposed DMFANet surpasses other SOTA methods in both effectiveness and efficiency.

In the future, we plan to explore lightweight network architectures to reduce the complexity of DMFANet. Additionally, we will develop an advanced framework integrating semi-supervised learning to alleviate the reliance on large-scale labeled datasets for model training.

REFERENCES

- [1] K. Lou, M. Li, F. Li, and X. Zheng, "Integrating local-global structural interaction using Siamese graph neural network for urban land use change detection from VHR satellite images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4413317.
- [2] J. Wang, H. Guo, X. Su, L. Zheng, and Q. Yuan, "PCDASNet: Position-constrained differential attention Siamese network for building damage assessment," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5622318.
- [3] Z. Lv et al., "Land cover change detection with heterogeneous remote sensing images: Review, progress, and perspective," *Proc. IEEE*, vol. 110, no. 12, pp. 1976–1991, Dec. 2022.
- [4] T. Bai et al., "Deep learning for change detection in remote sensing: A review," *Geo-Spatial Inf. Sci.*, vol. 26, no. 3, pp. 262–288, Jun. 2022.
- [5] Q. Shi, M. Liu, S. Li, X. Liu, F. Wang, and L. Zhang, "A deeply supervised attention metric-based network and an open aerial image dataset for remote sensing change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5604816.
- [6] G. Cheng et al., "Change detection methods for remote sensing in the last decade: A comprehensive review," *Remote Sens.*, vol. 16, no. 13, p. 2355, Jun. 2024.
- [7] Z. Liu, G. Li, G. Mercier, Y. He, and Q. Pan, "Change detection in heterogenous remote sensing images via homogeneous pixel transformation," *IEEE Trans. Image Process.*, vol. 27, no. 4, pp. 1822–1834, Apr. 2018.
- [8] A. Shafique, G. Cao, Z. Khan, M. Asad, and M. Aslam, "Deep learning-based change detection in remote sensing images: A review," *Remote Sens.*, vol. 14, no. 4, p. 871, Feb. 2022.
- [9] L. Wan, Y. Xiang, and H. You, "A post-classification comparison method for SAR and optical images change detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 7, pp. 1026–1030, Jul. 2019.
- [10] H. Zhang et al., "Panoramic change analysis framework (PCA-F): A new method for large-scale change detection in high-resolution remote sensing images," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2024, pp. 10259–10263.
- [11] F. He, H. Chen, S. Yang, and Z. Guo, "A small sample-based multiclass change detection method using change vector analysis with adaptive weight Gaussian mixture model," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5626416.
- [12] T. Kattenborn, J. Leitloff, F. Schiefer, and S. Hinz, "Review on convolutional neural networks (CNN) in vegetation remote sensing," *ISPRS J. Photogramm. Remote Sens.*, vol. 173, pp. 24–49, Mar. 2021.
- [13] Z. Li, F. Liu, W. Yang, S. Peng, and J. Zhou, "A survey of convolutional neural networks: Analysis, applications, and prospects," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 12, pp. 6999–7019, Dec. 2022.
- [14] W. Shi, M. Zhang, R. Zhang, S. Chen, and Z. Zhan, "Change detection based on artificial intelligence: State-of-the-art and challenges," *Remote Sens.*, vol. 12, no. 10, p. 1688, May 2020.
- [15] Q. Li, M. Zhang, Z. Yang, Y. Yuan, and Q. Wang, "Edge-guided perceptual network for infrared small target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5643510.
- [16] X. Zhao, L. Wang, Y. Zhang, X. Han, M. Deveci, and M. Parmar, "A review of convolutional neural networks in computer vision," *Artif. Intell. Rev.*, vol. 57, no. 4, p. 99, Mar. 2024.
- [17] R. C. Daudt, B. L. Saux, and A. Boulch, "Fully convolutional Siamese networks for change detection," in *Proc. 25th IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2018, pp. 4063–4067.
- [18] S. Li and L. Huo, "Remote sensing image change detection based on fully convolutional network with pyramid attention," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2021, pp. 4352–4355.
- [19] J. Chen et al., "DASNet: Dual attentive fully convolutional Siamese networks for change detection in high-resolution satellite images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 1194–1206, 2021.
- [20] H. Chen and Z. Shi, "A spatial-temporal attention-based method and a new dataset for remote sensing image change detection," *Remote Sens.*, vol. 12, no. 10, p. 1662, May 2020.
- [21] S. Fang, K. Li, J. Shao, and Z. Li, "SNUNet-CD: A densely connected Siamese network for change detection of VHR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.
- [22] W. G. C. Bandara and V. M. Patel, "A transformer-based Siamese network for change detection," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2022, pp. 207–210.
- [23] C. Zhang, L. Wang, S. Cheng, and Y. Li, "SwinSUNet: Pure transformer network for remote sensing image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5224713.
- [24] Q. Li, R. Zhong, X. Du, and Y. Du, "TransUNetCD: A hybrid transformer network for change detection in optical remote-sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5622519.
- [25] Y. Huang, X. Li, Z. Du, and H. Shen, "Spatiotemporal enhancement and interlevel fusion network for remote sensing images change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5609414.
- [26] Q. Hou, D. Zhou, and J. Feng, "Coordinate attention for efficient mobile network design," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 13708–13717.
- [27] Y. Feng, J. Jiang, H. Xu, and J. Zheng, "Change detection on remote sensing images using dual-branch multilevel intertemporal network," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4401015.
- [28] M. Sandler, A. Howard, M. Zhu, A. Zhmoginov, and L.-C. Chen, "MobileNetV2: Inverted residuals and linear bottlenecks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 4510–4520.
- [29] J. Ding, X. Li, J. Li, and S. Chen, "Multiple Spatial-Spectral features aggregated neural network for hyperspectral change detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.
- [30] S. Wang, W. Wu, Z. Zheng, and J. Li, "CTST: CNN and transformer-based spatio-temporally synchronized network for remote sensing change detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 16272–16288, 2024.
- [31] N. He, L. Wang, P. Zheng, C. Zhang, and L. Li, "CBSASNet: A Siamese network based on channel bias split attention for remote sensing change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5630117.
- [32] J. Ma, J. Duan, X. Tang, X. Zhang, and L. Jiao, "EATDer: Edge-assisted adaptive transformer detector for remote sensing change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5602015.
- [33] Z. Li et al., "Stade-CDNet: Spatial-temporal attention with difference enhancement-based network for remote sensing image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024.
- [34] H. Chen, J. Song, C. Han, J. Xia, and N. Yokoya, "Change-Mamba: Remote sensing change detection with spatiotemporal state space model," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4409720.
- [35] A. Codegoni, G. Lombardi, and A. Ferrari, "TINYCD: A (not so) deep learning model for change detection," *Neural. Comput. Appl.*, vol. 35, no. 11, pp. 8471–8486, Dec. 2022.
- [36] T. Lei et al., "Difference enhancement and spatial-spectral nonlocal network for change detection in VHR remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4507013.
- [37] T. Lei et al., "Ultralightweight spatial-spectral feature cooperation network for change detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4402114.
- [38] Z. Mao, Z. Luo, and Y. Tang, "Remote sensing building change detection with global high-frequency cues guidance and result-aware alignment," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.
- [39] M. Zhang, Q. Li, Y. Miao, Y. Yuan, and Q. Wang, "Difference-guided aggregation network with multiimage pixel contrast for change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5611114.
- [40] S. Holail, T. Saleh, X. Xiao, and D. Li, "AFDE-Net: Building change detection using attention-based feature differential enhancement for satellite imagery," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.
- [41] S. Pang, J. Lan, Z. Zuo, and J. Chen, "SFGT-CD: Semantic feature-guided building change detection from bitemporal remote-sensing images with transformers," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.
- [42] D. Lu, S. Cheng, L. Wang, and S. Song, "Multi-scale feature progressive fusion network for remote sensing image change detection," *Sci. Rep.*, vol. 12, no. 1, pp. 1026–1030, Jul. 2022.

- [43] F. Luo, T. Zhou, J. Liu, T. Guo, X. Gong, and J. Ren, "Multiscale diff-changed feature fusion network for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5502713.
- [44] Z. Li, C. Tang, L. Wang, and A. Y. Zomaya, "Remote sensing change detection via temporal feature interaction and guided refinement," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5628711.
- [45] X. Zhu, H. Hu, S. Lin, and J. Dai, "Deformable ConvNets v2: More deformable, better results," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 9308–9316.
- [46] C. Li, Y. Fan, and X. Cai, "PyConvU-net: A lightweight and multiscale network for biomedical image segmentation," *BMC Bioinf.*, vol. 22, no. 1, pp. 1–11, Jan. 2021.
- [47] Y. Gong, X. Yu, Y. Ding, X. Peng, J. Zhao, and Z. Han, "Effective fusion factor in FPN for tiny object detection," in *Proc. IEEE Winter Conf. Appl. Comput. Vis. (WACV)*, Jan. 2021, pp. 1159–1167.
- [48] S. Woo, J. Park, J.-Y. Lee, and I. S. Kweon, "CBAM: Convolutional block attention module," in *Proc. ECCV*, Munich, Germany, Sep. 2018, pp. 3–19.
- [49] X. Lin, L. Ma, W. Liu, and S.-F. Chang, "Context-gated convolution," in *Proc. Eur. Conf. Comput. Vis.*, Dec. 2020, pp. 701–718.
- [50] J. Yang et al., "Focal self-attention for local-global interactions in vision transformers," 2021, *arXiv:2107.00641*.
- [51] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2980–2988.
- [52] Z. Li et al., "Lightweight remote sensing change detection with progressive feature aggregation and supervised attention," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5602812.
- [53] S. Ji, S. Wei, and M. Lu, "Fully convolutional networks for multisource building extraction from an open aerial and satellite imagery data set," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 574–586, Jan. 2019.
- [54] C. Zhang et al., "A deeply supervised image fusion network for change detection in high resolution bi-temporal remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 166, pp. 183–200, Aug. 2020.
- [55] H. Zheng et al., "HFA-Net: High frequency attention Siamese network for building change detection in VHR remote sensing images," *Pattern Recognit.*, vol. 129, Sep. 2022, Art. no. 108717.
- [56] H. Chen, Z. Qi, and Z. Shi, "Remote sensing image change detection with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5607514.
- [57] H. Fu et al., "Three-dimensional singular spectrum analysis for precise land cover classification from UAV-borne hyperspectral benchmark datasets," *ISPRS J. Photogramm. Remote Sens.*, vol. 203, pp. 115–134, Sep. 2023.
- [58] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-CAM: Visual explanations from deep networks via gradient-based localization," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 618–626.



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